

genetics



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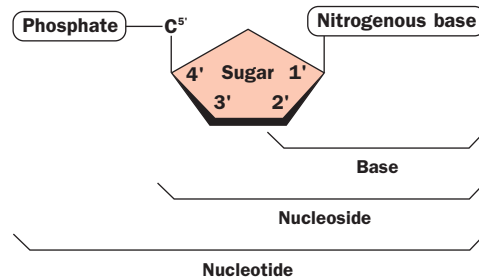
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For Your Reference

The following section provides a group of diagrams and illustrations applicable to many entries in this encyclopedia. The molecular structures of DNA and RNA are provided in detail in several different formats, to help the student understand the structures and visualize how these molecules combine and interact. The full set of human chromosomes are presented diagrammatically, each of which is shown with a representative few of the hundreds or thousands of genes it carries.

NUCLEOTIDE STRUCTURE



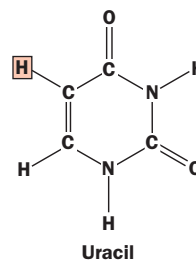
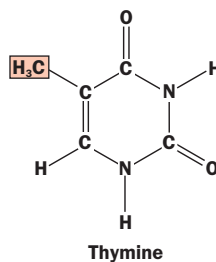
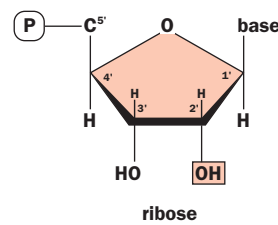
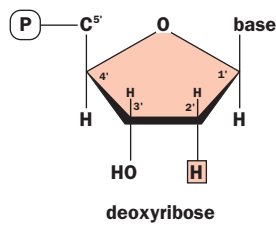
Sample naming conventions for each structure:

Adenine

Adenosine

Adenosine monophosphate

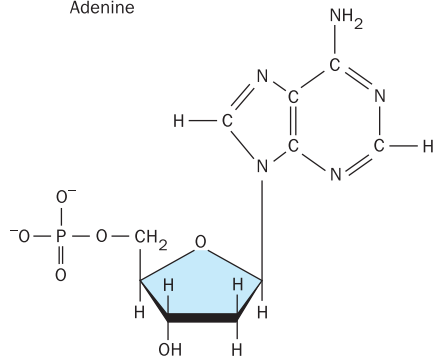
DNA VS. RNA



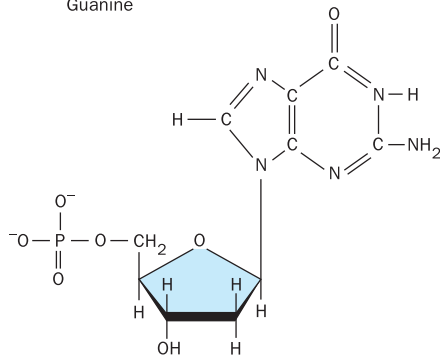
NUCLEOTIDE STRUCTURES

Purine-containing DNA nucleotides

Adenine

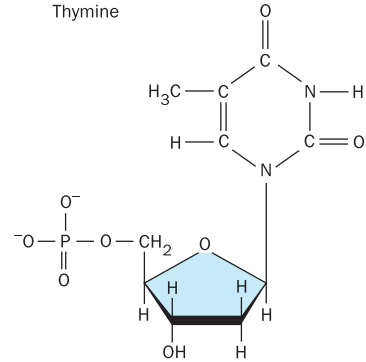


Guanine

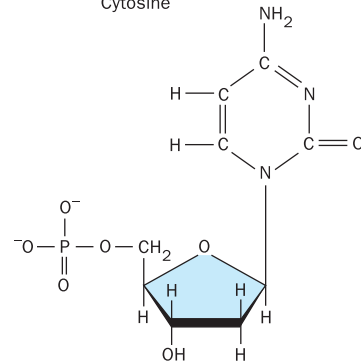


Pyrimidine-containing DNA nucleotides

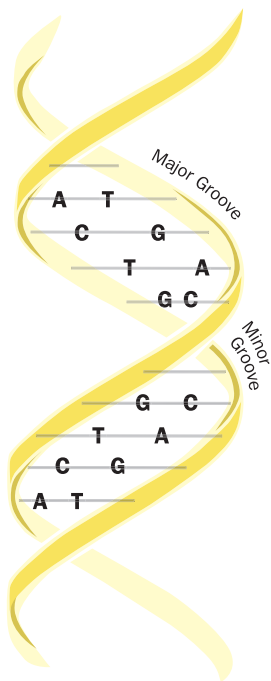
Thymine



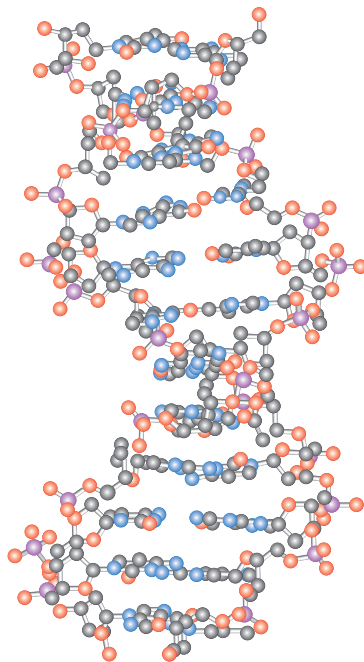
Cytosine



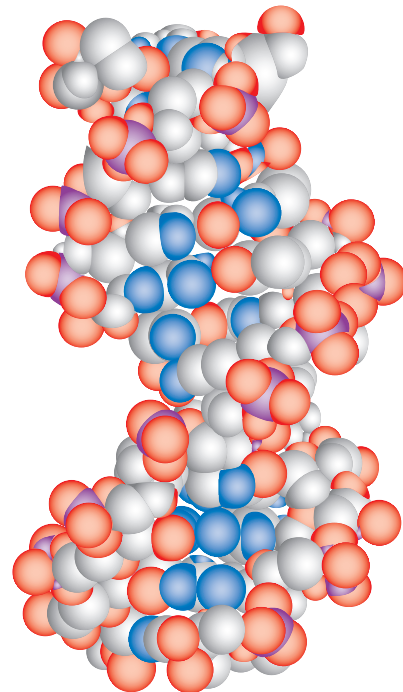
CANONICAL B-DNA DOUBLE HELIX



Ribbon model

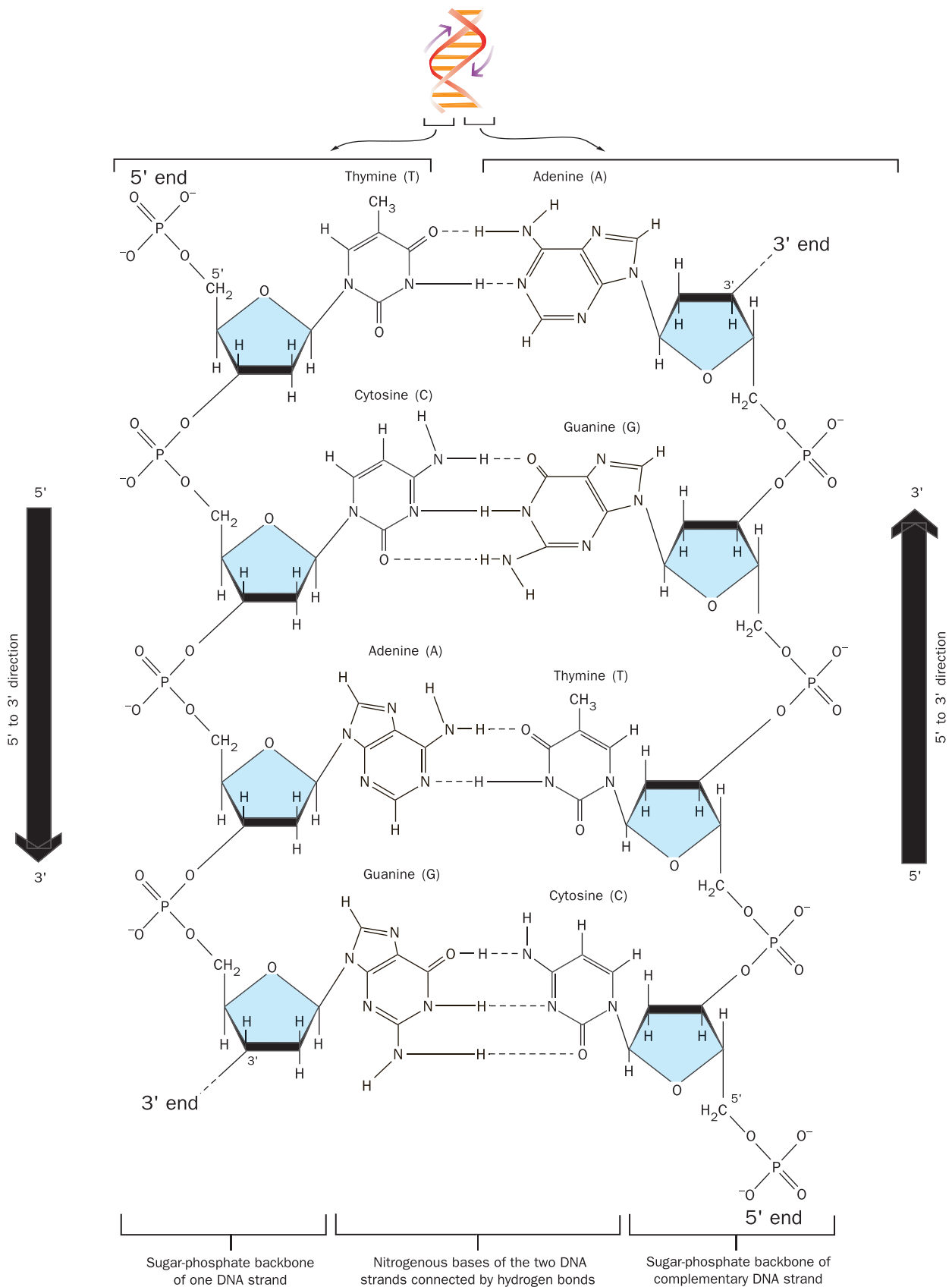


Ball-and-stick model

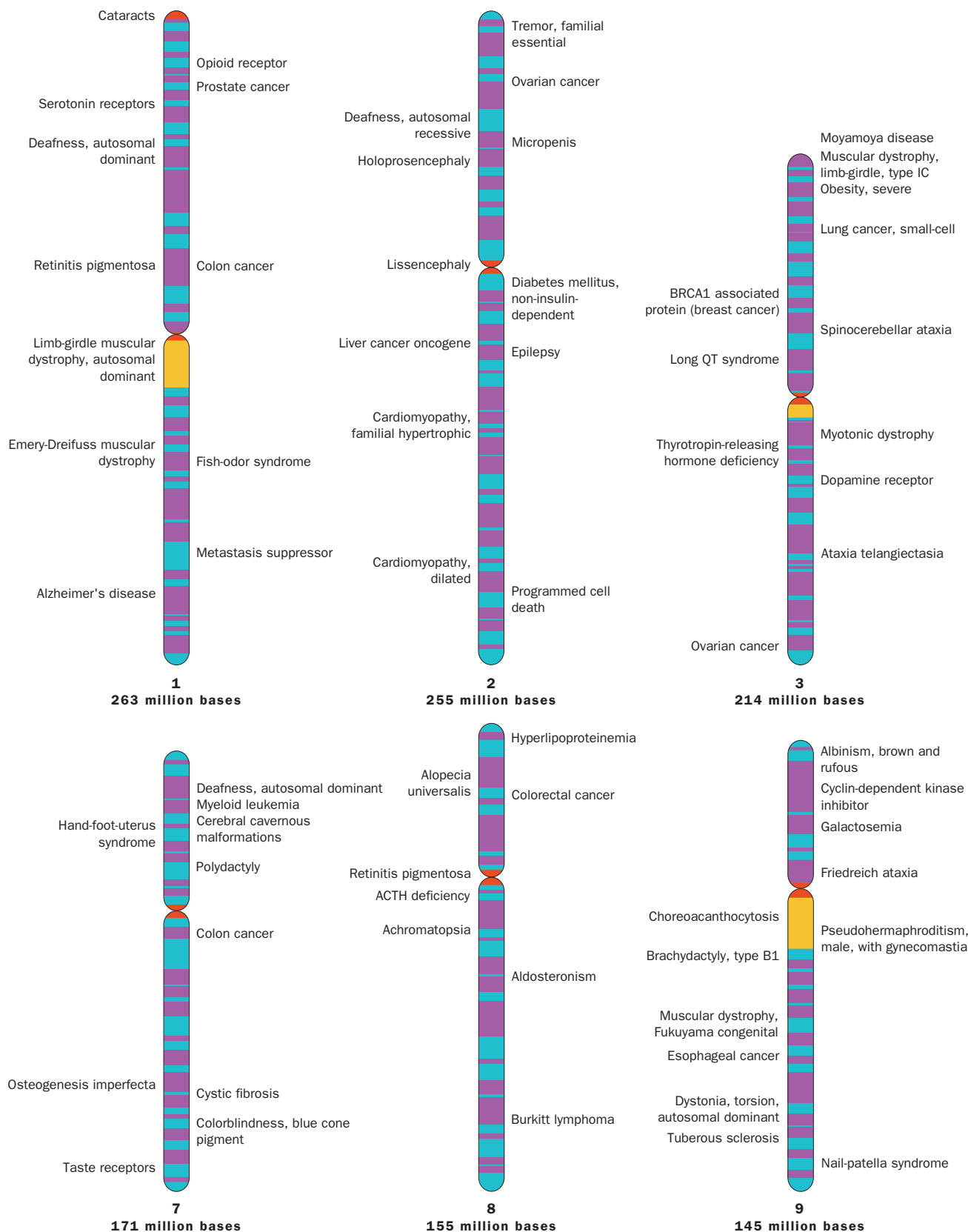


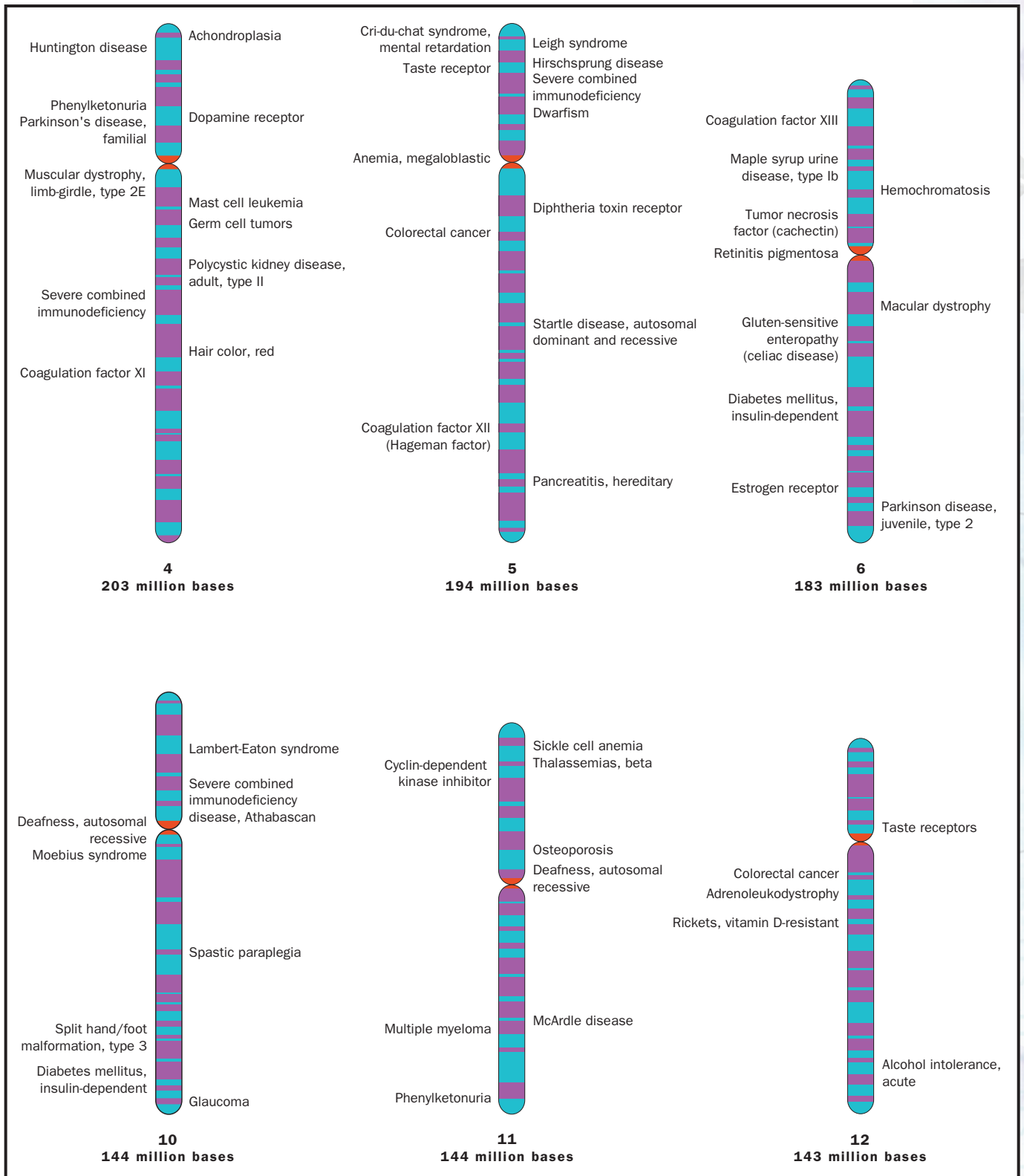
Space-filling model

DNA NUCLEOTIDES PAIR UP ACROSS THE DOUBLE HELIX; THE TWO STRANDS RUN ANTI-PARALLEL

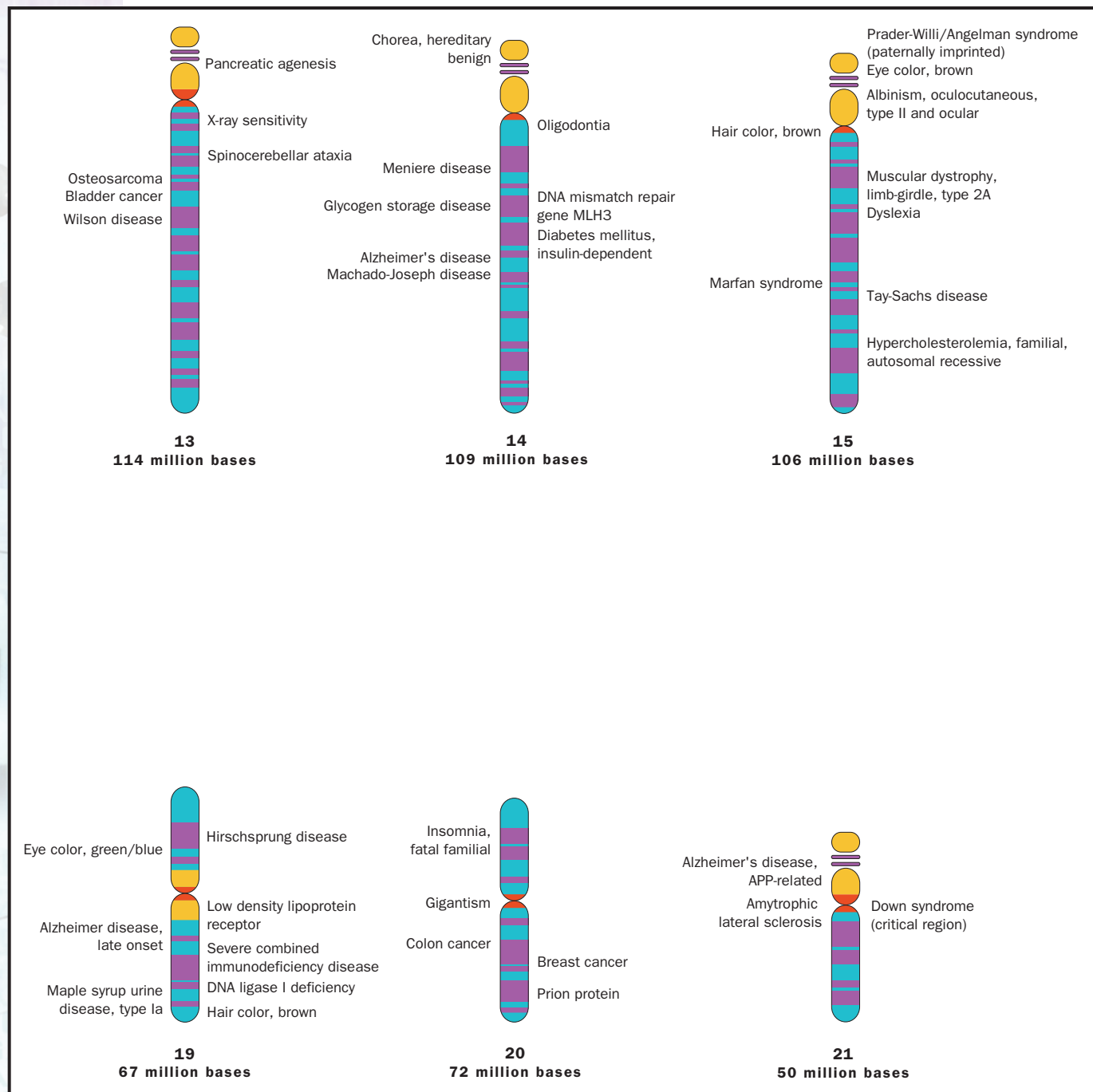


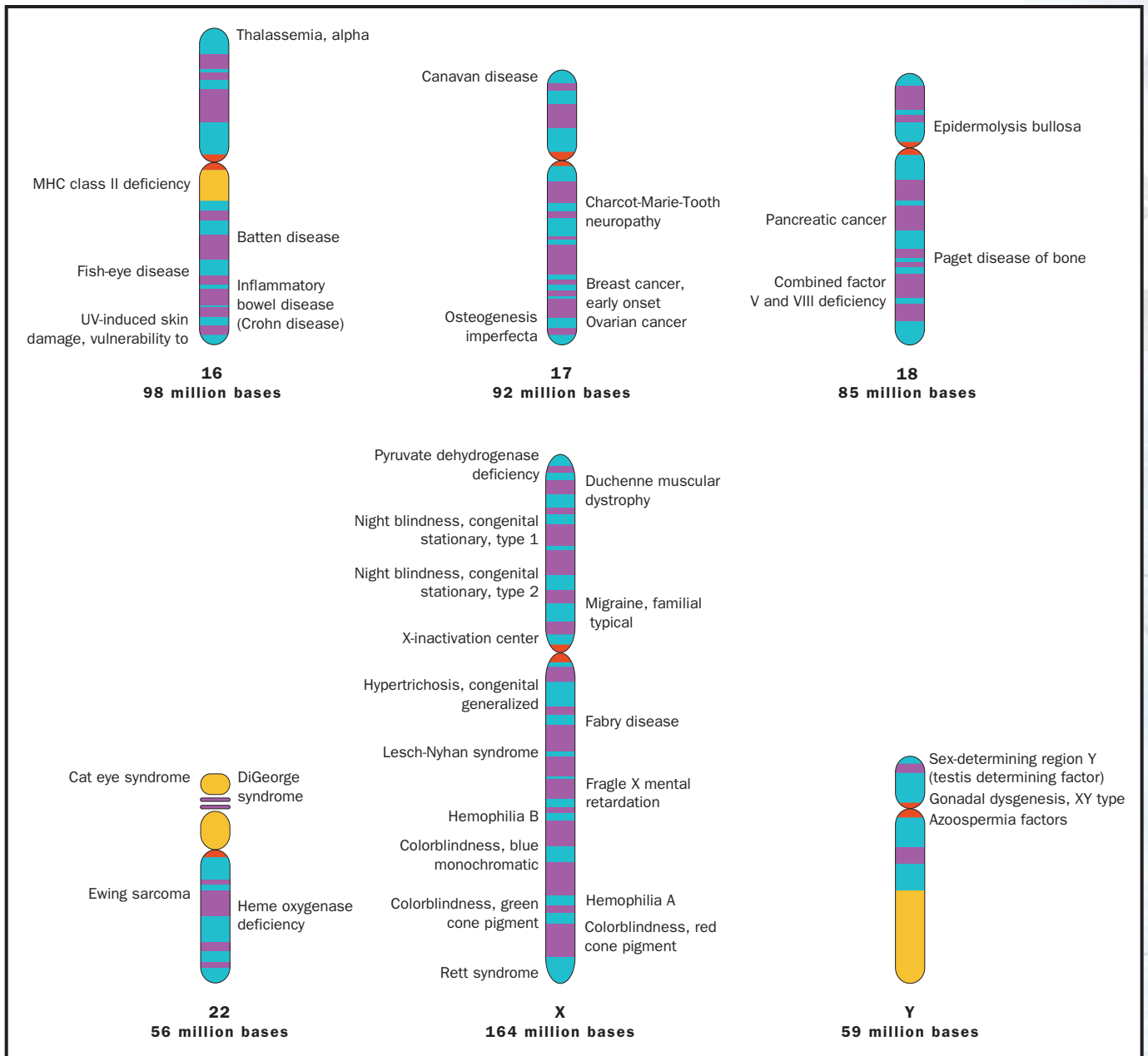
SELECTED LANDMARKS OF THE HUMAN GENOME





For Your Reference





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Table of Contents

VOLUME 1

PREFACE	v
FOR YOUR REFERENCE	ix
LIST OF CONTRIBUTORS	xvii

A

Accelerated Aging: Progeria	1
Addiction	4
Aging and Life Span	6
Agricultural Biotechnology	9
Alternative Splicing	11
Alzheimer's Disease	14
Ames Test	19
Androgen Insensitivity Syndrome	21
Antibiotic Resistance	26
Antisense Nucleotides	29
Apoptosis	31
<i>Arabidopsis thaliana</i>	33
Archaea	36
Attention Deficit Hyperactivity Disorder	39
Attorney	42
Automated Sequencer	43

B

Balanced Polymorphism	45
Behavior	46
Bioinformatics	52
Biopesticides	57
Bioremediation	59
Biotechnology	62
Biotechnology Entrepreneur	65
Biotechnology: Ethical Issues	66
Biotechnology and Genetic Engineering, History	70
Birth Defects	74

Blood Type	82
Blotting	86
Breast Cancer	89

C

Cancer	92
Carcinogens	97
Cardiovascular Disease	101
Cell Cycle	103
Cell, Eukaryotic	108
Centromere	114
Chaperones	116
Chromosomal Aberrations	119
Chromosomal Banding	125
Chromosomal Theory of Inheritance, History	129
Chromosome, Eukaryotic	132
Chromosome, Prokaryotic	139
Chromosomes, Artificial	144
Classical Hybrid Genetics	146
Clinical Geneticist	149
Cloning Genes	152
Cloning: Ethical Issues	158
Cloning Organisms	161
College Professor	165
Colon Cancer	166
Color Vision	170
Combinatorial Chemistry	173
Complex Traits	177
Computational Biologist	181
Conjugation	182
Conservation Biology: Genetic Approaches	186
Conservation Geneticist	190
Crick, Francis	192
Crossing Over	194



Cycle Sequencing	198
Cystic Fibrosis	199

D

Delbrück, Max	203
Development, Genetic Control of	204
Diabetes	209
Disease, Genetics of	213
DNA	215
DNA Footprinting	220
DNA Libraries	222
DNA Microarrays	225
DNA Polymerases	230
DNA Profiling	233
DNA Repair	239
DNA Structure and Function, History	248
DNA Vaccines	253
Down Syndrome	256
PHOTO CREDITS	259
GLOSSARY	263
TOPICAL OUTLINE	281
INDEX	287

VOLUME 2

FOR YOUR REFERENCE	v
LIST OF CONTRIBUTORS	xiii

E

Educator	1
Embryonic Stem Cells	3
Epidemiologist	6
Epistasis	7
<i>Escherichia coli</i> (<i>E. coli</i> bacterium)	9
Eubacteria	11
Eugenics	16
Evolution, Molecular	21
Evolution of Genes	26
Eye Color	31

F

Fertilization	33
Founder Effect	36
Fragile X Syndrome	39
Fruit Fly: <i>Drosophila</i>	42

G

Gel Electrophoresis	45
---------------------------	----

Gene	50
Gene and Environment	54
Gene Discovery	57
Gene Expression: Overview of Control ..	61
Gene Families	67
Gene Flow	70
Gene Targeting	71
Gene Therapy	74
Gene Therapy: Ethical Issues	80
Genetic Code	83
Genetic Counseling	87
Genetic Counselor	91
Genetic Discrimination	92
Genetic Drift	94
Genetic Testing	96
Genetic Testing: Ethical Issues	101
Genetically Modified Foods	106
Geneticist	110
Genetics	111
Genome	112
Genomic Medicine	118
Genomics	120
Genomics Industry	123
Genotype and Phenotype	125
Growth Disorders	129

H

Hardy-Weinberg Equilibrium	133
Hemoglobinopathies	136
Hemophilia	141
Heterozygote Advantage	146
High-Throughput Screening	149
HIV	150
Homology	156
Hormonal Regulation	158
HPLC: High-Performance Liquid Chromatography	165
Human Disease Genes, Identification of ..	167
Human Genome Project	171
Human Immunodeficiency Virus	178
Huntington's Disease	178
Hybrid Superiority	178

I

Immune System Genetics	178
Imprinting	183
<i>In situ</i> Hybridization	186
Inbreeding	189

Individual Genetic Variation	191
Information Systems Manager	192
Inheritance, Extranuclear	194
Inheritance Patterns	199
Intelligence	207
Internet	211
PHOTO CREDITS	215
GLOSSARY	219
TOPICAL OUTLINE	237
INDEX	243

VOLUME 3

FOR YOUR REFERENCE	v
LIST OF CONTRIBUTORS	xiii

L

Laboratory Technician	1
Legal Issues	3
Linkage and Recombination	4

M

Maize	8
Mapping	11
Marker Systems	15
Mass Spectrometry	18
McClintock, Barbara	21
McKusick, Victor	22
Meiosis	24
Mendel, Gregor	30
Mendelian Genetics	32
Metabolic Disease	37
Methylation	46
Microbiologist	50
Mitochondrial Diseases	51
Mitochondrial Genome	55
Mitosis	57
Model Organisms	60
Molecular Anthropology	62
Molecular Biologist	70
Morgan, Thomas Hunt	72
Mosaicism	76
Muller, Hermann	80
Multiple Alleles	82
Muscular Dystrophy	83
Mutagen	87
Mutagenesis	89

Mutation	93
Mutation Rate	98

N

Nature of the Gene, History	101
Nomenclature	106
Nondisjunction	108
Nucleases	112
Nucleotide	115
Nucleus	119

O

Oncogenes	127
Operon	131
Overlapping Genes	135

P

Patenting Genes	136
Pedigree	138
Pharmaceutical Scientist	142
Pharmacogenetics and Pharmacogenomics	144
Physician Scientist	147
Plant Genetic Engineer	149
Plasmid	150
Pleiotropy	153
Polymerase Chain Reaction	154
Polymorphisms	159
Polyploidy	163
Population Bottleneck	167
Population Genetics	171
Population Screening	175
Post-translational Control	178
Prenatal Diagnosis	182
Prion	187
Privacy	190
Probability	193
Protein Sequencing	196
Proteins	198
Proteomics	205
Pseudogenes	209
Psychiatric Disorders	213
Public Health, Genetic Techniques in ..	216
Purification of DNA	220
PHOTO CREDITS	223
GLOSSARY	227
TOPICAL OUTLINE	245
INDEX	251



VOLUME 4

FOR YOUR REFERENCE	v
LIST OF CONTRIBUTORS	xiii

Q

Quantitative Traits	1
---------------------------	---

R

Reading Frame	4
Recombinant DNA	5
Repetitive DNA Sequences	7
Replication	12
Reproductive Technology	19
Reproductive Technology: Ethical Issues ..	26
Restriction Enzymes	31
Retrovirus	34
Reverse Transcriptase	39
Ribosome	42
Ribozyme	44
RNA	46
RNA Interference	54
RNA Processing	57
Rodent Models	60
Roundworm: <i>Caenorhabditis elegans</i>	62

S

Sanger, Fred	64
Science Writer	65
Selection	67
Sequencing DNA	69
Severe Combined Immune Deficiency ...	74
Sex Determination	78
Sexual Orientation	83
Signal Transduction	85
Speciation	91
Statistical Geneticist	93
Statistics	95

T

Tay-Sachs Disease	98
Technical Writer	102
Telomere	104
Transcription	106
Transcription Factors	112
Transduction	117
Transformation	121
Transgenic Animals	124
Transgenic Microorganisms	127
Transgenic Organisms: Ethical Issues ..	129
Transgenic Plants	132
Translation	135
Transplantation	139
Transposable Genetic Elements	143
Triplet Repeat Disease	148
Tumor Suppressor Genes	153
Twins	155

V

Viroids and Virusoids	162
Virus	164

W

Watson, James	171
---------------------	-----

X

X Chromosome	173
--------------------	-----

Y

Y Chromosome	176
Yeast	179

Z

Zebrafish	181
-----------------	-----

PHOTO CREDITS	185
---------------------	-----

GLOSSARY	189
----------------	-----

TOPICAL OUTLINE	207
-----------------------	-----

CUMULATIVE INDEX	213
------------------------	-----

Quantitative Traits

Quantitative traits are those that vary continuously. This is in contrast to qualitative traits, in which the **phenotype** is discrete and can take on one of only a few different values. Examples of quantitative traits include height, weight, and blood pressure. There is no single gene for any of these traits, instead it is generally believed that continuous variation in a trait such as blood pressure is partly due to DNA sequence variations at multiple genes, or **loci**. Such loci are referred to as quantitative trait loci (QTL). Much of how we study and characterize quantitative traits can be attributed to the work of Ronald Fisher and Sewall Wright, accomplished during the first half of the twentieth century.

The Genetic Architecture of Quantitative Traits

An important goal of genetic studies is to characterize the genetic architecture of quantitative traits. Genetic architecture can be defined in one of four ways. First, it refers to the number of QTLs that influence a quantitative trait. Second, it can mean the number of **alleles** that each QTL has. Third, it reflects the frequencies of the alleles in the population. And fourth, it refers to the influence of each QTL and its alleles on the quantitative trait. Imagine, for instance, a quantitative trait influenced by 6 loci, each of which has 3 alleles. This gives a total of 18 possible allele combinations. Some alleles may be very rare in a population, so that the phenotypes it contributes to may be rare as well. Some alleles have disproportionate effects on the phenotype (for instance, an allele that causes dwarfism), which may mask the more subtle effects of other alleles. The trait may also be influenced by the environment, giving an even wider range of phenotypic possibilities.

Understanding the genetic architecture of quantitative traits is important in a number of disciplines, including animal and plant breeding, medicine, and evolution. For example, a quantitative trait of interest to animal breeders might be meat quality in pigs. The identification and characterization of QTLs for meat quality might provide a basis for selecting and breeding pigs with certain desirable features. In medicine, an important goal is to identify genetic risk factors for various common diseases. Many genetic studies of common disease focus on the presence or absence of disease as the trait of interest. In some cases, however, quantitative traits may provide more information for identifying genes than qualitative traits. For example,



phenotype observable characteristics of an organism

loci site on a chromosome (singular, locus)

alleles particular forms of genes



Human height is a quantitative trait, controlled by multiple genes. The broad distribution of heights reflects this fact. Note that most people have an intermediate height, a typical distribution pattern for quantitative traits.

genotypes sets of genes present

identifying genetic risk factors for cardiovascular disease might be facilitated by studying the genetic architecture of cholesterol metabolism or blood pressure rather than the presence or absence of cardiovascular disease itself. Cholesterol metabolism is an example of an intermediate trait or endophenotype for cardiovascular disease. That is, it is related to the disease and may be useful as a “proxy measure” of the disease.

QTLs and Complex Effects on Phenotype

It is important to note that QTLs can influence quantitative traits in a number of different ways. First, variation at a QTL can impact quantitative trait levels. That is, the average or mean of the observed phenotypes for the trait may be different among different **genotypes** (for example, some genotypes will produce taller organisms than others). This is important because much of the basic theory underlying statistical methods for studying quantitative traits is based on genotypic means. For this reason, most genetic studies focus on quantitative trait means. However, there are a variety of other ways QTLs can influence quantitative traits. For example, it is possible that the trait means are the same among different genotypes but that the variances (the spread on either side of the mean) are not. In other words, variation in phenotypic values may be greater for some genotypes than for others—some genotypes, for example, may give a wider range of heights than others. This is believed to be due to gene-gene and gene-environment interactions such that the magnitude of the effects of a particular environmental or genetic factor may differ across genotypes.

It is also possible for QTLs to influence the relationship or correlation among quantitative traits. For example, the rate at which two proteins bind might be due to variation in the QTLs that code for those proteins. As a final example, QTLs can also impact the dynamics of a trait. That is, change in a phenotype over time might be due to variation at a

QTL, such as when blood pressure varies with the age of the individual. Thus, QTLs can affect quantitative trait levels, variability, co-variability, and dynamics.

In addition, each type of QTL effect may depend on a particular genetic or environmental context. Thus, the influence of a particular QTL on quantitative trait levels, variability, covariability, or dynamics may depend on one or more other QTLs (an effect called **epistasis** or gene-gene interaction) and/or one or more environmental factors. Although such context-dependent effects may be very common, and may play an important role in genetic architecture, they are typically very difficult to detect and characterize. This is partly due to limits of available statistical methods and the availability of large sample sizes.

Analysis of Quantitative Traits

Characterization of the genetic architecture of quantitative traits is typically carried out using one of two different study designs. The first approach starts with the quantitative trait of interest (such as height or blood pressure) and attempts to draw inferences about the underlying genetics from looking at the degree of trait resemblance among related subjects. This approach is sometimes referred to as a top-down or unmeasured genotype strategy because the inheritance pattern of the trait is the focus and no genetic variations are actually measured. The top-down approach is often the first step taken to determine whether there is evidence for a genetic component.

Heritability (the likelihood that the trait will be passed on to offspring) and **segregation analysis** are examples of statistical analyses that use a top-down approach. With the bottom-up or measured genotype approach, candidate QTLs are measured and then used to draw inferences about which genes might play a role in the genetic architecture of a quantitative trait. Prior to the availability of technologies for measuring QTLs, the top-down approach was very common. However, it is now inexpensive and efficient to measure many QTLs, making the bottom-up strategy a common study design. **Linkage analysis** and **association analysis** are two general statistical approaches that utilize the bottom-up study design.

The definition and characterization of quantitative traits is changing very rapidly. New technologies such as DNA microarrays and protein mass spectrometry are making it possible to quantitatively measure the expression levels of thousands of genes simultaneously. These new measures make it possible to study gene expression at both the RNA level and the protein level as a quantitative trait. These new quantitative traits open the door for understanding the hierarchy of the relationship between QTL variation and variation in quantitative traits at both the biochemical and physiological level. SEE ALSO COMPLEX TRAITS; DNA MICROARRAYS; GENE DISCOVERY; LINKAGE AND RECOMBINATION.

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epistasis suppression of a characteristic of one gene by the action of another gene

segregation analysis statistical test to determine pattern of inheritance for a trait

linkage analysis examination of co-inheritance of disease and DNA markers, used to locate disease genes

association analysis estimation of the relationship between alleles or genotypes and disease



codon a sequence of three mRNA nucleotides coding for one amino acid

transcription messenger RNA formation from a DNA sequence

glycolipid molecule composed of sugar and fatty acid

Insertion of two Cs shifts the reading frame, creating a premature stop codon.

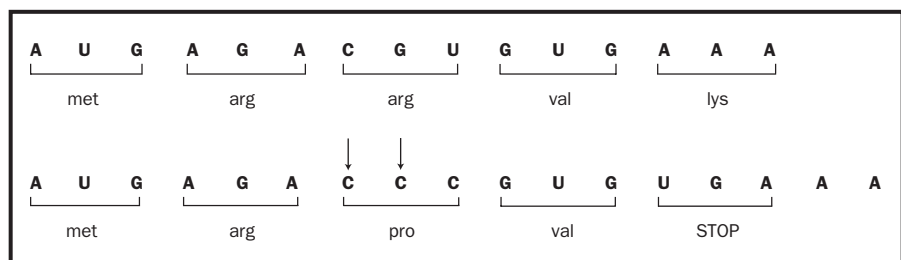
Reading Frame

Almost all organisms translate their genes into protein structures using an identical, universal **codon** dictionary in which each amino acid in the protein is represented by a combination of only three nucleotides. For example, the sequence AAA in a gene is transcribed into the sequence UUU in messenger RNA (mRNA) and is then translated as the amino acid phenylalanine. A group of several codons that, taken together, provide the code for an amino acid, is called a reading frame. There are no “spaces” in the mRNA to denote the end of one codon and the start of another. Instead, the reading frame, or group of triplets, is determined solely by initial position of the pattern-making machinery at the start of the translation. In order for correct translation to occur, this reading frame must be maintained throughout the **transcription** and translation process.

Any single or double base insertions or deletions in the DNA or RNA sequence will throw off the reading frame and result in aberrant gene expression. Mutations that result in such insertions or deletions are termed “frameshift mutations.” The insertion of three nucleotides, on the other hand, will only extend the length of the protein without affecting the reading frame, although it may affect the function of the protein. Several genetic diseases, including Huntington’s disease, contain such trinucleotide repeats.

Because DNA consists of four possible bases and each codon consists of only a three-base sequence, there are 4^3 , or sixty-four possible codons for the twenty common amino acids. In the codon dictionary, sixty-one of the codons code for amino acids, with the remaining three codons marking the end of the reading frame. The codon AUG denotes both the amino acid methionine and the start of the reading frame. In several cases, more than one codon can result in the creation of the same amino acid. For example CAC and CAU both code for histidine. This condition is termed “degeneracy,” and it means that some mutations may still result in the same amino acid being inserted at that point into the protein. The above example also explains the “wobble hypothesis,” put forward by Francis Crick, which states that substitutions in the terminal nucleotide of a codon have little or no effect on the proper insertion of amino acids during translation.

Medically important frameshift mutations include an insertion in the gene for a rare form of Gaucher disease preventing **glycolipid** breakdown. Charcot-Marie-Tooth disease, which results in numbness in hands and feet, is caused by the repetitive insertion of 1.5 million base pairs into the gene. A frameshift mutation of four bases in the gene coding for the low-density lipid receptor near one end causes the receptor to improperly anchor itself in the cell membrane, resulting in the faulty turnover of cholesterol that



causes hypercholesterolemia, or high blood levels of cholesterol. A single nucleotide pair deletion in codon 55 of the gene coding for phenylalanine hydroxylase (PAH) results in a form of phenylketonuria. Frameshift mutations are denoted by listing the location and specific change in the DNA. For example, 55delT indicates a thymidine was deleted in the 55th codon of the PAH gene. SEE ALSO CRICK, FRANCIS; GENETIC CODE; MUTATION; TRANSCRIPTION; TRANSLATION.

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Recessiveness See *Inheritance Patterns*

Recombinant DNA

Recombinant DNA refers to a collection of techniques for creating (and analyzing) DNA molecules that contain DNA from two unrelated organisms. One of the DNA molecules is typically a bacterial or viral DNA that is capable of accepting another DNA molecule; this is called a **vector** DNA. The other DNA molecule is from an organism of interest, which could be anything from a bacterium to a whale, or a human. Combining these two DNA molecules allows for the **replication** of many copies of a specific DNA. These copies of DNA can be studied in detail, used to produce valuable proteins, or used for gene therapy or other applications.

The development of recombinant DNA tools and techniques in the early 1970s led to much concern about developing genetically modified organisms with unanticipated and potentially dangerous properties. This concern led to a proposal for a voluntary moratorium on recombinant DNA research in 1974, and to a meeting in 1975 at the Asilomar Conference Center in California. Participants at the Asilomar Conference agreed to a set of safety standards for recombinant DNA work, including the use of disabled bacteria that were unable to survive outside the laboratory. This conference helped satisfy the public about the safety of **recombinant DNA** research, and led to a rapid expansion of the use of these powerful new technologies.

Overview of Recombination Techniques

The basic technique of recombinant DNA involves digesting a vector DNA with a **restriction enzyme**, which is a molecular scissors that cuts DNA at specific sites. A DNA molecule from the organism of interest is also digested, in a separate tube, with the same restriction enzyme. The two DNAs are then mixed together and joined, this time using an enzyme called DNA ligase, to make an intact, double-stranded DNA molecule. This construct is

A U G C G A U C C C C C

Three different reading frames for one mRNA sequence

vector carrier

replication duplication of DNA

recombinant DNA DNA formed by combining segments of DNA, usually from different types of organisms

restriction enzyme an enzyme that cuts DNA at a particular sequence



genome the total genetic material in a cell or organism

then put into *Escherichia coli* cells, where the resulting DNA is copied billions of times. This novel DNA molecule is then isolated from the *E. coli* cells and analyzed to make sure that the correct construct was produced. This DNA can then be sequenced, used to generate protein from *E. coli* or another host, or for many other purposes.

There are many variations on this basic method of producing recombinant DNA molecules. For example, sometimes researchers are interested in isolating a whole collection of DNAs from an organism. In this case, they digest the whole **genome** with restriction enzyme, join many DNA fragments into many different vector molecules, and then transform those molecules into *E. coli*. The different *E. coli* cells that contain different DNA molecules are then pooled, resulting in a “library” of *E. coli* cells that contain, collectively, all of the genes present in the original organism.

Another variation is to make a library of all expressed genes (genes that are used to make proteins) from an organism or tissue. In this case, RNA is isolated. The isolated RNA is converted to DNA using the enzyme called reverse transcriptase. The resulting DNA copy, commonly abbreviated as cDNA, is then joined to vector molecules and put into *E. coli*. This collection of recombinant cDNAs (a cDNA library) allows researchers to study the expressed genes in an organism, independent from nonexpressed DNA.

Applications

Recombinant DNA technology has been used for many purposes. The Human Genome Project has relied on recombinant DNA technology to generate libraries of genomic DNA molecules. Proteins for the treatment or diagnosis of disease have been produced using recombinant DNA techniques. In recent years, a number of crops have been modified using these methods as well.

As of 2001, over eighty products that are currently used for treatment of disease or for vaccination had been produced using recombinant DNA techniques. The first was human insulin, which was produced in 1978. Other protein therapies that have been produced using recombinant DNA technology include hepatitis B vaccine, human growth hormone, clotting factors for treating hemophilia, and many other drugs. At least 350 additional recombinant-based drugs are currently being tested for safety and efficacy. In addition, a number of diagnostic tests for diseases, including tests for hepatitis and AIDS, have been produced with recombinant DNA technology.

Gene therapy is another area of applied genetics that requires recombinant DNA techniques. In this case, the recombinant DNA molecules themselves are used for therapy. Gene therapy is being developed or attempted for a number of inherited human diseases.

Recombinant DNA technology has also been used to produce genetically modified foods. These include tomatoes that can be vine-ripened before shipping and rice with improved nutritional qualities. Genetically modified foods have generated controversy, and there is an ongoing debate in some communities about the benefits and risks of developing crops using recombinant DNA technology.

Since the mid-1970s, recombinant DNA techniques have been widely applied in research laboratories and in pharmaceutical and agricultural companies. It is likely that this relatively new area of genetics will continue to play an increasingly important part in biological research into the foreseeable future. SEE ALSO BIOTECHNOLOGY; CLONING GENES; CROSSING OVER; DNA LIBRARIES; *ESCHERICHIA COLI*; GENE THERAPY: ETHICAL ISSUES; GENETICALLY MODIFIED FOODS; HUMAN GENOME PROJECT; PLASMID; RESTRICTION ENZYMES; REVERSE TRANSCRIPTASE; TRANSPOSABLE GENETIC ELEMENTS.

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Repetitive DNA Elements

The human genome contains approximately three billion **base pairs** of DNA. Within this there are between 30,000 and 70,000 genes, which together add up to less than 5 percent of the entire **genome**. Most of the rest is made up of several types of noncoding repeated elements.

Most gene sequences are unique, found only once in the genome. In contrast, repetitive DNA elements are found in multiple copies, in some cases thousands of copies, as shown in Table 1. Unlike genes, most repetitive elements do not code for protein or RNA. Repetitive elements have been found in most other eukaryotic genomes that have been analyzed. What functions they serve, if any, are mainly unknown. Their presence and spread causes several inherited diseases, and they have been linked to major events in evolution.

Types of Repetitive Elements

Repetitive elements differ in their position in the genome, sequence, size, number of copies, and presence or absence of coding regions within them. The two major classes of repetitive elements are interspersed elements and tandem arrays.

Interspersed repeated elements are usually present as single copies and distributed widely throughout the genome. The interspersed repeats alone constitute about 45 percent of the genome. The best-characterized interspersed repeats are the transposable genetic elements, also called mobile elements or “jumping genes” (Figure 1).

base pairs two nucleotides (either DNA or RNA) linked by weak bonds

genome the total genetic material in a cell or organism

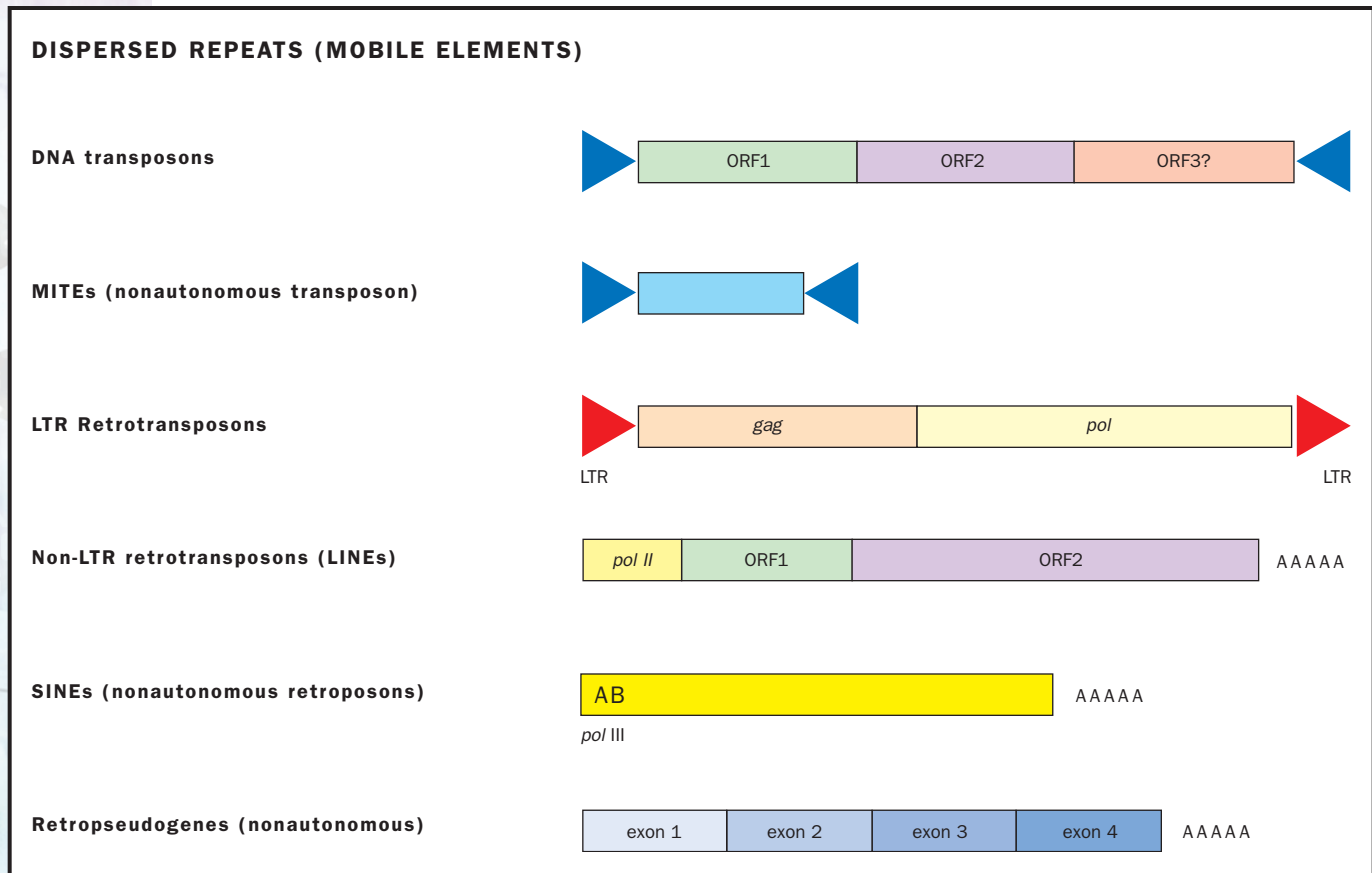


Figure 1. Dispersed repeats are usually found as single units spread throughout the genome. The DNA transposons have two or three open reading frames (ORFs) coding for the factors needed for amplification. LTR retrotransposons are flanked by long terminal repeats (LTR) and have genes such as *gag* (group antigen gene), and *pol* (polymerase gene). Non-LTR retrotransposons or LINEs will have ORFs and are transcribed by RNA polymerase II (*pol II*). There are two nonautonomous repeats thought to parasitize LINEs for its amplification: SINEs and retropseudogenes.

Sequences that are “tandemly arrayed” are present as duplicates, either head to tail or head to head. So-called satellites, minisatellites, and microsatellites largely exist in the form of tandem arrays (these elements originally got their name as “satellites” because they separated from the bulk of nuclear DNA during centrifugation). Sequences repeated in tandem are common at the centromere (where the two halves of a replicated chromosome are held together), and at or near the telomeres (the chromosome tips). Because they are difficult to sequence, sequences repeated in tandem at centromeres and telomeres are underrepresented in the draft sequence of the human genome. This makes it difficult to estimate the copy number, but they certainly represent at least 10 percent of the genome.

Tandem Arrays

Satellites (also called classical satellites), which occur in four classes (I–IV), form arrays of 1,000 to 10 million repeated units, particularly in the **heterochromatin** of chromosomes. They are concentrated in centromeres and account for much of the DNA there. Satellites of one type, called alpha-satellites, occur as repeated units of approximately 171 base pairs (bp) in length, with high levels of sequence variation between the repeated units, as shown in Figure 2.

heterochromatin condensed portion of chromosomes

NUMBER OF COPIES OF INTERSPERSED REPEATS OBSERVED IN THE DRAFT OF THE HUMAN GENOME

Repeat type	Copy number	Fraction of the genome (%)
DNA transposons	294	3
LTR retrotransposons	443	8
Non-LTR retrotransposons:	2,426	34
LINEs	868	21
SINEs	1,558	13
Others	3	0.1

Table 1.

Minisatellites form arrays of several hundred units of 7 to 100 bp in length. They are present everywhere with an increasing concentration toward the telomeres. They differ from satellites in that they are found only in moderate numbers of tandem repeats and because of their high degree of dispersion throughout chromosomes.

Microsatellites, or simple sequence repeats (SSRs), are composed of units of one to six **nucleotides**, repeated up to a length of 100 bp or more. One-third are simple “polyadenylated” repeats, composed of nothing but adenine nucleotides. Other examples of abundant microsatellites are $(AC)_n$, $(AAAN)_n$, $(AAAA)_n$, and $(AAN)_n$, where N represents any nucleotide and n is the number of repeats. Less abundant, but important because of their direct involvement in the generation of disease, are the $(CAG/CTG)_n$ and $(CGG/CCG)_n$ trinucleotide (or triplet) repeats.

Telomeric and subtelomeric repeats are present at the end of the telomeres and are composed of short tandem repeats (STRs) of $(TTAGGG)_n$, up to 30,000 bp long. This sequence is “highly conserved,” meaning it has changed very little over evolutionary time, indicating it likely plays a very important role. These STRs function as caps or ends of the long linear chromosomal DNA molecule and are crucial to the maintenance of intact eukaryotic chromosomes. Subtelomeric repeats act as transitions between the boundary of the telomere and the rest of the chromosome. They contain units similar to the TTAGGG, but they are not conserved.

nucleotides the building blocks of RNA or DNA

Transposable Elements

Transposable elements are classified as either transposons or retrotransposons, depending on their mechanism of amplification. Transposons directly synthesize a DNA copy of themselves, whereas retrotransposons generate an RNA intermediate that is then reverse-transcribed (by the enzyme reverse transcriptase) back into DNA. Transposable elements fall into three major groups: DNA transposons, long terminal repeat (LTR) retrotransposons, and non-LTR retrotransposons. They also are subdivided into autonomous and nonautonomous elements, based on whether they can move independently within the genome or require other elements to perform this process, as shown in Figure 3.

DNA transposons are flanked by inverted repeats and contain two or more open reading frames (ORFs). An ORF is a DNA sequence that can be transcribed to make protein. The ORFs in DNA transposons code for the proteins required for making transposon copies and spreading them through the genome. The nonautonomous elements miniature inverted-repeat transposable elements (MITEs) are derived from a parent DNA transposon that

Figure 2. Schematic organization of the structures characteristic of one type of repeated sequences found in eukaryotic genomes. Tandem repeats consist of a different variety of sequence units (shown in parentheses) that are found repeated side by side, with the number of copies being very high.

TANDEM REPEATS	
Satellites	I (42 bp) _n
	II, III (ATTCC or other) _n
	alpha (171 bp) _n
Mini satellites and microsatellites	(7-100 bp) _n
	(1-6 bp) _n
Telomeric and subtelomeric repeats	(TTAGGG) _n
bp = base pairs	
n = number of times the unit within parenthesis repeats	

lost ORF sequences, making them unable to amplify on their own. Instead, they must borrow the factors for amplification from external sources.

LTR retrotransposons are very similar to the genomes of retroviruses. They are flanked by 250 to 600 bp direct repeats called long terminal repeats. In general, not only are these elements defective, but they also appear to have deletions typical of nonautonomous families.

Several different groups of non-LTR retrotransposons can be found throughout most, if not all, eukaryotic genomes. One of these groups, the long interspersed repeated elements (LINEs), constitute about 21 percent of the human genome, with L1 and L2 being the dominant elements. Most of the element copies are incomplete and inactive. Two types of non-autonomous elements are thought to use factors made by LINEs: short interspersed repeated elements (SINEs) and retropseudogenes.

SINEs are derived from two types of genes coding for RNA: 7SL (which aids the movement of new proteins into the **endoplasmic reticulum**) and transfer RNAs. The most abundant human SINE is Alu, constituting about 13 percent of the human genome.

Retropseudogenes are derived from retrotransposition of **mRNA** derived from different genes. They can be distinguished from the parental gene by their lack of a functional **promoter** and by their lack of **introns**. The human genome is estimated to contain 35,000 copies of different retropseudogenes.

Role of Repetitive DNA in Evolution and Impact on the Human Genome

Most eukaryotic genomes contain repetitive DNA. Although most repeated sequences have no known function, their impact and importance on genomes is evident. Mobile repeated elements have been a critical factor in gene evolution. It has been suggested that some types of repeats may be linked to speciation, since during the evolutionary period when there was a high activity of mobile elements, radiation of different species occurred.

endoplasmic reticulum
network of membranes within the cell

mRNA messenger RNA

promoter DNA sequence to which RNA polymerase binds to begin transcription

introns untranslated portions of genes that interrupt coding regions

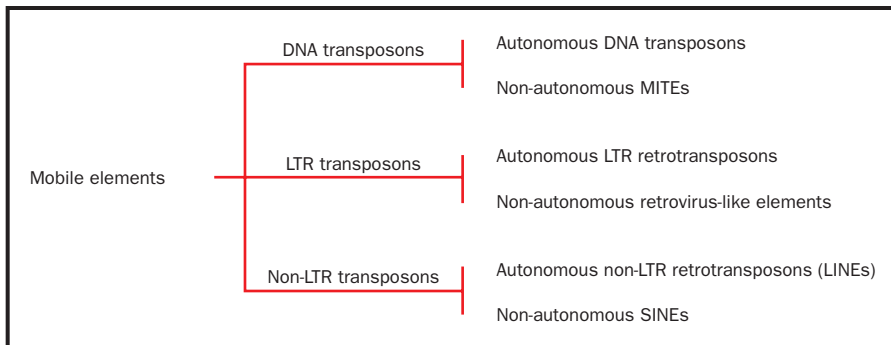


Figure 3. Representation of the division of the major types of mobile elements. Mobile elements can be subdivided into groups depending on their mechanism of amplification (DNA or RNA based), their structure (presence of LTR elements) and their ability for autonomous and nonautonomous amplification.

There are several diseases linked to—or caused by—repetitive elements. Expansion of triplet repeats has been tied to fragile X syndrome (a common cause of mental retardation), Huntington’s disease, myotonic muscular dystrophy, and several other diseases. In addition, the discovery of STR instability in certain cancers suggests that sequence instability may play a role in cancer progression.

Mobile elements have caused diseases when a new mobile element disrupts an important gene. Neurofibromatosis type 1, for example, is caused by the insertion of an Alu element in the gene *NF1*. Alternatively, recombination between two repeated elements within a gene will alter its function, also causing disease. Many examples of cancers (e.g., acute myelogenous leukemia) and inherited diseases (e.g., alpha thalassemia) are caused by mobile-element-based recombinations.

Application of Repeats to Human Genomic Studies

Repeated sequences can be useful genetic tools. Because many of the repeated sequences are stably inherited, highly conserved, and found throughout the genome, they are ideal for genetic studies: They can act as “signposts” for finding and mapping functional genes. In addition, a repeat at a particular locus may be absent in one individual, or it may differ between two individuals (**polymorphism**). This makes repeats useful for identifying specific individuals (called DNA profiling) and their ancestors (molecular anthropology).

Microsatellites, in particular, have been used to identify individuals, study populations, and construct evolutionary trees. They have also been used as markers for disease-gene mapping and to evaluate specific genes in tumors. LINEs, and particularly the human SINE Alu, have been used for studies of human population genetics, primate comparative genomics, and DNA profiling. **SEE ALSO** CENTROMERE; CHROMOSOME, EUKARYOTIC; *IN SITU* HYBRIDIZATION; POLYMORPHISMS; PSEUDOGENES; RETROVIRUS; TELOMERE; TRANSPOSABLE GENETIC ELEMENTS; TRIPLET REPEAT DISEASE.

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polymorphism DNA sequence variant